WEEK 1 — PYTHON FUNDAMENTALS

1. Introduction to Python and its Advantages
2. PEP Standards
    - PEP 8
    - Naming conventions
    - Code formatting
    - Pythonic code
3. Python Datatypes & Built-in Functions
    - Primitive Datatypes
        - Integer
        - String
        - Float
        - Boolean
    - List
    - Tuple
    - Dictionary
    - Set
    - Type conversion
    - Mutable vs Immutable types
4. Typing
    - Type annotations
    - str
    - int
    - float

    - bool
    - list
    - dict
    - tuple
    - set
    - Optional
    - Union
    - Any
    - Type aliases
5. Operators
    - Arithmetic Operators
    - Comparison / Relational Operators
    - Assignment Operators
    - Logical Operators
    - Bitwise Operators
    - Membership Operators (in, not in)
    - Identity Operators (is, is not)
    - Walrus Operator (:=)
6. Conditional Flow Control
    - if
    - if-else
    - if-elif-else
    - Nested conditions
    - while
    - for

    - break
    - continue
    - pass
7. Comprehensions
    - List Comprehension
    - Set Comprehension
    - Dictionary Comprehension
    - Generator Expression
    - Conditional Comprehensions
    - Nested Comprehensions
8. Generators
    - yield
    - Generator functions
    - Generator expressions
    - Lazy evaluation
    - next()
    - Generator vs List
9. Python Functions
    - Defining functions
    - Return values
    - Positional Arguments
    - Keyword Arguments
    - Default Arguments
    - *args
    - **kwargs

- Recursion
- Anonymous Functions (lambda)
- Local and Global Scope
- global
- nonlocal

10. Built-in Functions
    - map()
    - zip()
    - filter()
    - reduce()
    - enumerate()
    - type()
    - id()
    - range()
    - sorted()
    - max()
    - min()

WEEK 2 — PYTHON DATA MODELING, OOP & INTERNALS

1. Collections
    - Counter
    - OrderedDict
    - defaultdict
    - namedtuple

    - deque
2. Dataclass
    - @dataclass
    - Fields and Type Annotations
    - Default Values
    - field()
    - default_factory
    - Frozen Dataclasses
    - Dataclass vs Normal Class
    - Dataclass vs NamedTuple
3. Enum / IntEnum
    - Enum
    - Enum Members
    - Enum Values
    - auto()
    - IntEnum
    - Comparing Enum Values
    - Practical Use Cases
4. Python Memory Management & Garbage Collection
    - Variables and Object References
    - Object Identity
    - Reference Counting
    - Garbage Collection
    - Circular References
    - id()

- gc Module
- Mutable vs Immutable Objects

5. Pydantic
    - Introduction to Pydantic
    - Pydantic Models
    - Field Validation
    - Type Validation
    - Default Values
    - Optional Fields
    - Nested Models
    - Serialization
    - Deserialization
    - Pydantic vs Dataclass
6. Object-Oriented Programming

    6.1 Class & Objects
    - Classes
    - Objects
    - Constructors
    - Instance Attributes
    - Class Attributes
    - Instance Methods
    - Class Methods
    - Static Methods

    6.2 OOP Concepts

    - Inheritance
    - Encapsulation
    - Polymorphism
    - Abstract Classes
    - Abstract Methods
7. Dunder / Magic Methods
    - init
    - str
    - repr
    - len
    - eq
    - lt
    - add
    - getitem
    - setitem
    - iter
    - next
    - call
8. Descriptors
    - What are Descriptors?
    - get
    - set
    - delete
    - property
    - Descriptor Protocol

- Practical Descriptor Examples

WEEK 3 — ADVANCED PYTHON

1. Decorators
    - Functions as First-Class Objects
    - Higher-Order Functions
    - Basic Decorators
    - Decorators with Arguments
    - *args and **kwargs in Decorators
    - Multiple Decorators
    - functools.wraps
    - Practical Use Cases
2. Monkey Patching
    - What is Monkey Patching?
    - Runtime Modification
    - Patching Functions
    - Patching Classes
    - Patching Objects
    - Patching Modules
    - Monkey Patching in Testing
    - unittest.mock
    - patch()
    - Risks and Best Practices
3. Exception Handling
    - Exceptions vs Errors

    - try
    - except
    - else
    - finally
    - Multiple Exceptions
    - Custom Exceptions
    - raise
    - Exception Chaining
    - Best Practices
4. Multithreading
    - Threads
    - threading Module
    - Thread Creation
    - Thread Synchronization
    - Locks
    - Race Conditions
    - Thread Pool
5. Multiprocessing
    - Processes vs Threads
    - multiprocessing
    - Process Creation
    - Process Pool
    - Inter-Process Communication
6. GIL — Global Interpreter Lock
    - What is GIL?

- Why Python Has GIL
- GIL and Threads
- CPU-Bound vs I/O-Bound Tasks
- Threads vs Processes

7. Async / Await
    - Synchronous vs Asynchronous Programming
    - async
    - await
    - Coroutines
    - Event Loop
    - asyncio
    - Async I/O
    - Async vs Threading vs Multiprocessing
8. Logging
    - Why Logging?
    - Logging Levels
        - DEBUG
        - INFO
        - WARNING
        - ERROR
        - CRITICAL
    - Log Formatting
    - File Logging
    - logging Module
    - Logging Best Practices

9. Unit Testing
    - Why Unit Testing?
    - Test Cases
    - unittest
    - Assertions
    - Test Fixtures
    - setUp()
    - tearDown()
    - Mocking
    - Patching
    - Testing Exceptions
    - Test Organization
10. Regular Expressions
    - Regex Basics
    - Character Classes
    - Quantifiers
    - Groups
    - Anchors
    - match()
    - search()
    - findall()
    - sub()
    - split()
    - Practical Validation Examples
11. File Handling

- Opening Files
- Reading Files
- Writing Files
- Append Mode
- Context Managers
- with
- Text Files
- CSV
- JSON
- File Paths

FASTAPI & SQLALCHEMY

1. InProject - FastAPI Upskilling plan
2. CDP - FastAPI Upskilling plan